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RESULTS

Results

01 · MULTIPLE LINEAR REGRESSION

one numeric outcome, several predictors

Table 1. Regression coefficients

	(1)	β	VIF
(Intercept)	20.36 (6.03) [8.11, 32.62]		
pre_score	0.61 (0.08) [0.46, 0.77]	—	1.25
age	0.02 (0.07) [-0.12, 0.16]	—	1.41
group: b	5.35 (1.64) [2.01, 8.69]	—	1.04
method: online	-2.05 (2.02) [-6.16, 2.06]	—	1.25
method: workshop	-3.25 (2.27) [-7.87, 1.37]	—	1.25
Num.Obs.	40		
R ²	0.75		
R ² Adj.	0.71		
F	20.42		
RMSE	4.68		
AIC	250.96		
BIC	262.78		
Log.Lik.	-118.48		

assumption checks: linearity, normality, homoscedasticity, and multicollinearity (VIF).

Figure. Residual diagnostics

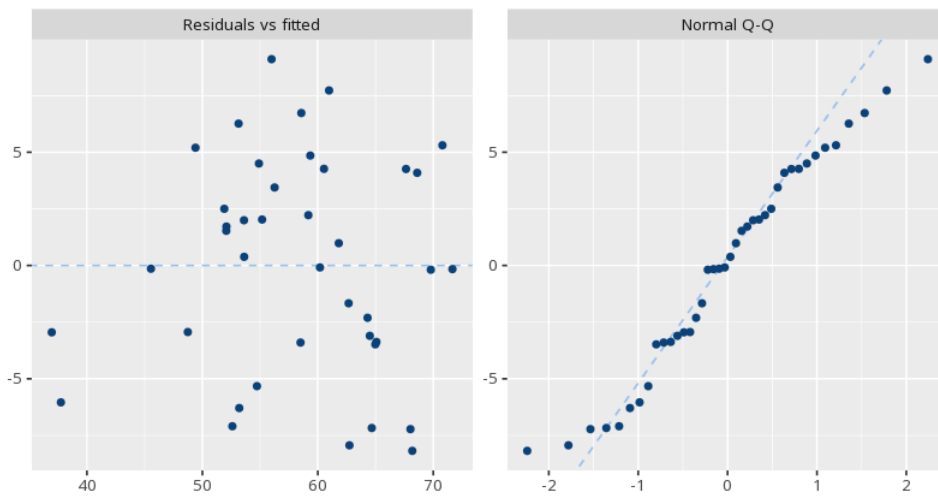
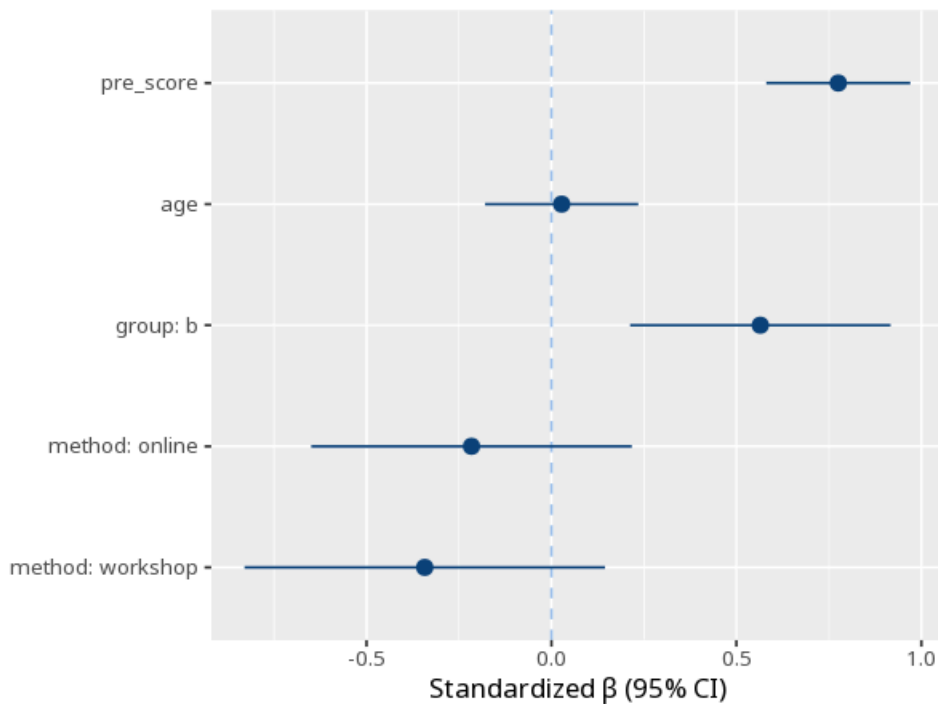


Figure. Coefficient plot



Understanding the numbers

R². The share of the outcome's variance jointly explained by all predictors. Here, $R^2 = .75$, so the model accounts for about 75% of the variance in the outcome.

VIF. How much a predictor's variance is inflated by overlap with the other predictors (multicollinearity); values above ~5-10 are cause for concern (O'Brien, 2007). Here, the largest VIF in this run is 1.41.

RMSE. The root-mean-square error: the typical size of the model's prediction error, in the outcome's own units. Here, $RMSE = 4.68$.

R² Adj.. R^2 , adjusted for the number of predictors in the model. Here, adjusted $R^2 = 0.71$.

AIC. A fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here, $AIC = 250.96$.

BIC. Like AIC but penalizes complexity more heavily as sample size grows. Here, BIC = 262.78.

Log.Lik.. The log-likelihood: how probable the observed data are under the fitted model; it underlies AIC and BIC. Here, Log.Lik. = -118.48.

N. The number of observations the model was fit on. Here, N = 40.

F. The overall F-test of whether the model explains more variance than a model with no predictors at all. Here, F = 20.42.

B. Each predictor's effect holding the others constant, in that predictor's own units. Here, predictor pre_score gave B = 0.61.

How to read this test

Each predictor's B is its effect holding the others constant; p and CI assess it. R² is the joint variance explained; VIF flags predictors that overlap too much (multicollinearity). B is in each predictor's own units, so to compare which predictors matter most use the standardized coefficient (β), not B. Your significance threshold (α) is 0.05.

APA template: The model explained R²=.75 of the variance, F(5,34)=20.42, p < .001; predictor pre_score gave B=0.61, p < .001.

Statistical basis: Ordinary least squares regression coefficients (B), SE, CI (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Standardized coefficients (β) (Lüdtke D, Ben-Shachar MS, Patil I, Makowski D (2020). "Extracting, Computing and Exploring the Parameters of Statistical Models using R." Journal of Open Source Software, 5(53), 2445.); Variance inflation factor (VIF) cutoff guidance (O'Brien, R. M. (2007). "A caution regarding rules of thumb for variance inflation factors." Quality & Quantity, 41(5), 673-690.). Full references in the exported CITATIONS.txt.

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